```
mysql> use Banking_managment;
Database changed
mysql> CREATE TABLE users (
    ->     user_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     password_hash VARCHAR(255) NOT NULL,
    ->     role ENUM('CUSTOMER','ADMIN') DEFAULT 'CUSTOMER',
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> SELECT *FROM Banking_managment;
ERROR 1146 (42S02): Table 'banking_managment.banking_managment' doesn't exist
mysql> SELECT *FROM USERS;
Empty set (0.01 sec)

mysql> CREATE TABLE accounts (
    ->     account_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     user_id INT NOT NULL,
    ->     account_number VARCHAR(20) UNIQUE NOT NULL,
    ->     account_type ENUM('SAVINGS','CURRENT') NOT NULL,
    ->     balance DECIMAL(15,2) DEFAULT 0.00,
    ->     status ENUM('ACTIVE','BLOCKED') DEFAULT 'ACTIVE',
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_users_accounts FOREIGN KEY(user_id)
    ->     REFERENCES users(user_id)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE transactions (
    ->     transaction_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     from_account INT,
    ->     to_account INT,
    ->     amount DECIMAL(15,2) NOT NULL,
    ->     transaction_type ENUM('DEPOSIT','WITHDRAWAL','TRANSFER') NOT NULL,
    ->     status ENUM('SUCCESS','FAILED') DEFAULT 'SUCCESS',
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_from FOREIGN KEY(from_account) REFERENCES accounts(account_id),
    ->     CONSTRAINT fk_to FOREIGN KEY(to_account) REFERENCES accounts(account_id)
    -> );
```

```
mysql> CREATE TABLE audit_log (
    ->     log_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     user_id INT,
    ->     action VARCHAR(255) NOT NULL,
    ->     ip_address VARCHAR(45),
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_users_audit FOREIGN KEY(user_id)
    ->     REFERENCES users(user_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO users (name, email, password_hash, role) VALUES
    -> ('Aarav Kumar', 'aarav@gmail.com', 'hash001', 'CUSTOMER'),
    -> ('Meera Sharma', 'meera@gmail.com', 'hash002', 'CUSTOMER'),
    -> ('Rohan Verma', 'rohan@gmail.com', 'hash003', 'CUSTOMER'),
    -> ('Ananya Iyer', 'ananya@gmail.com', 'hash004', 'CUSTOMER'),
    -> ('Vikram Singh', 'vikram@gmail.com', 'hash005', 'CUSTOMER'),
    -> ('Priya Nair', 'priya@gmail.com', 'hash006', 'CUSTOMER'),
    -> ('Kiran Patel', 'kiran@gmail.com', 'hash007', 'CUSTOMER'),
    -> ('Neha Reddy', 'neha@gmail.com', 'hash008', 'CUSTOMER'),
    -> ('Sanjay Rao', 'sanjay@gmail.com', 'hash009', 'CUSTOMER'),
    -> ('Admin User', 'admin@bank.com', 'adminhash', 'ADMIN');
Query OK, 10 rows affected (0.02 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> INSERT INTO accounts (user_id, account_number, account_type, balance, status) VALUES
    -> (1, 'ACC10001', 'SAVINGS', 45000.00, 'ACTIVE'),
    -> (2, 'ACC10002', 'SAVINGS', 12000.00, 'ACTIVE'),
    -> (3, 'ACC10003', 'CURRENT', 80000.00, 'ACTIVE'),
    -> (4, 'ACC10004', 'SAVINGS', 25000.00, 'ACTIVE'),
    -> (5, 'ACC10005', 'CURRENT', 9500.00, 'ACTIVE'),
    -> (6, 'ACC10006', 'SAVINGS', 33000.00, 'ACTIVE'),
    -> (7, 'ACC10007', 'SAVINGS', 67000.00, 'ACTIVE'),
    -> (8, 'ACC10008', 'CURRENT', 15000.00, 'ACTIVE'),
    -> (9, 'ACC10009', 'SAVINGS', 5000.00, 'BLOCKED'),
    -> (10,'ACC10010', 'CURRENT', 99999.00, 'ACTIVE');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO transactions (from_account, to_account, amount, transaction_type, status) VALUES
    -> (NULL, 1, 5000.00, 'DEPOSIT', 'SUCCESS'),
    -> (1, NULL, 2000.00, 'WITHDRAWAL', 'SUCCESS'),
    -> (2, 3, 1500.00, 'TRANSFER', 'SUCCESS'),
    -> (3, 4, 7000.00, 'TRANSFER', 'SUCCESS'),
    -> (NULL, 5, 12000.00, 'DEPOSIT', 'SUCCESS'),
    -> (6, NULL, 3000.00, 'WITHDRAWAL', 'SUCCESS'),
    -> (7, 8, 2500.00, 'TRANSFER', 'SUCCESS'),
    -> (NULL, 9, 1000.00, 'DEPOSIT', 'FAILED'),
    -> (10, 1, 8500.00, 'TRANSFER', 'SUCCESS'),
    -> (4, NULL, 500.00, 'WITHDRAWAL', 'SUCCESS');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> INSERT INTO audit_log (user_id, action, ip_address) VALUES
    -> (1, 'Login successful', '192.168.0.10'),
    -> (2, 'Password updated', '192.168.0.11'),
    -> (3, 'Transfer initiated', '192.168.0.12'),
    -> (4, 'Deposit attempt', '192.168.0.13'),
    -> (5, 'Withdrawal successful', '192.168.0.14'),
    -> (6, 'Account status checked', '192.168.0.15'),
    -> (7, 'Transfer completed', '192.168.0.16'),
    -> (8, 'Login failed', '192.168.0.17'),
    -> (9, 'Blocked account access attempt', '192.168.0.18'),
    -> (10,'Admin audited transactions', '192.168.0.1');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SHOW TABLES;
+-----------------------------+
| Tables_in_banking_managment |
+-----------------------------+
| accounts                    |
| audit_log                   |
| transactions                |
| users                       |
+-----------------------------+
4 rows in set (0.02 sec)
```

```
mysql> select *from users;
+---------+--------------+------------------+---------------+----------+---------------------+
| user_id | name         | email            | password_hash | role     | created_at          |
+---------+--------------+------------------+---------------+----------+---------------------+
|       1 | Aarav Kumar  | aarav@gmail.com  | hash001       | CUSTOMER | 2026-01-27 09:23:21 |
|       2 | Meera Sharma | meera@gmail.com  | hash002       | CUSTOMER | 2026-01-27 09:23:21 |
|       3 | Rohan Verma  | rohan@gmail.com  | hash003       | CUSTOMER | 2026-01-27 09:23:21 |
|       4 | Ananya Iyer  | ananya@gmail.com | hash004       | CUSTOMER | 2026-01-27 09:23:21 |
|       5 | Vikram Singh | vikram@gmail.com | hash005       | CUSTOMER | 2026-01-27 09:23:21 |
|       6 | Priya Nair   | priya@gmail.com  | hash006       | CUSTOMER | 2026-01-27 09:23:21 |
|       7 | Kiran Patel  | kiran@gmail.com  | hash007       | CUSTOMER | 2026-01-27 09:23:21 |
|       8 | Neha Reddy   | neha@gmail.com   | hash008       | CUSTOMER | 2026-01-27 09:23:21 |
|       9 | Sanjay Rao   | sanjay@gmail.com | hash009       | CUSTOMER | 2026-01-27 09:23:21 |
|      10 | Admin User   | admin@bank.com   | adminhash     | ADMIN    | 2026-01-27 09:23:21 |
+---------+--------------+------------------+---------------+----------+---------------------+
10 rows in set (0.00 sec)

mysql> select *from Audit Logs;
ERROR 1146 (42S02): Table 'banking_managment.audit' doesn't exist
mysql> select *from audit_log;
+--------+---------+-------------------------------+--------------+---------------------+
| log_id | user_id | action                        | ip_address   | created_at          |
+--------+---------+-------------------------------+--------------+---------------------+
|      1 |       1 | Login successful              | 192.168.0.10 | 2026-01-27 09:24:01 |
|      2 |       2 | Password updated              | 192.168.0.11 | 2026-01-27 09:24:01 |
|      3 |       3 | Transfer initiated            | 192.168.0.12 | 2026-01-27 09:24:01 |
|      4 |       4 | Deposit attempt               | 192.168.0.13 | 2026-01-27 09:24:01 |
|      5 |       5 | Withdrawal successful         | 192.168.0.14 | 2026-01-27 09:24:01 |
|      6 |       6 | Account status checked        | 192.168.0.15 | 2026-01-27 09:24:01 |
|      7 |       7 | Transfer completed            | 192.168.0.16 | 2026-01-27 09:24:01 |
|      8 |       8 | Login failed                  | 192.168.0.17 | 2026-01-27 09:24:01 |
|      9 |       9 | Blocked account access attempt | 192.168.0.18 | 2026-01-27 09:24:01 |
|     10 |      10 | Admin audited transactions    | 192.168.0.1  | 2026-01-27 09:24:01 |
+--------+---------+-------------------------------+--------------+---------------------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT name, email
    -> FROM users
    -> WHERE role = 'CUSTOMER';
+--------------+------------------+
| name         | email            |
+--------------+------------------+
| Aarav Kumar  | aarav@gmail.com  |
| Meera Sharma | meera@gmail.com  |
| Rohan Verma  | rohan@gmail.com  |
| Ananya Iyer  | ananya@gmail.com |
| Vikram Singh | vikram@gmail.com |
| Priya Nair   | priya@gmail.com  |
| Kiran Patel  | kiran@gmail.com  |
| Neha Reddy   | neha@gmail.com   |
| Sanjay Rao   | sanjay@gmail.com |
+--------------+------------------+
9 rows in set (0.00 sec)

mysql> SELECT account_number, balance
    -> FROM accounts
    -> ORDER BY balance DESC;
+----------------+----------+
| account_number | balance  |
+----------------+----------+
| ACC10010       | 99999.00 |
| ACC10003       | 80000.00 |
| ACC10007       | 67000.00 |
| ACC10001       | 45000.00 |
| ACC10006       | 33000.00 |
| ACC10004       | 25000.00 |
| ACC10008       | 15000.00 |
| ACC10002       | 12000.00 |
| ACC10005       |  9500.00 |
| ACC10009       |  5000.00 |
+----------------+----------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT *FROM accounts ORDER BY balance DESC;
+------------+---------+----------------+--------------+----------+---------+---------------------+
| account_id | user_id | account_number | account_type | balance  | status  | created_at          |
+------------+---------+----------------+--------------+----------+---------+---------------------+
|         10 |      10 | ACC10010       | CURRENT      | 99999.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          3 |       3 | ACC10003       | CURRENT      | 80000.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          7 |       7 | ACC10007       | SAVINGS      | 67000.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          1 |       1 | ACC10001       | SAVINGS      | 45000.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          6 |       6 | ACC10006       | SAVINGS      | 33000.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          4 |       4 | ACC10004       | SAVINGS      | 25000.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          8 |       8 | ACC10008       | CURRENT      | 15000.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          2 |       2 | ACC10002       | SAVINGS      | 12000.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          5 |       5 | ACC10005       | CURRENT      |  9500.00 | ACTIVE  | 2026-01-27 09:23:40 |
|          9 |       9 | ACC10009       | SAVINGS      |  5000.00 | BLOCKED | 2026-01-27 09:23:40 |
+------------+---------+----------------+--------------+----------+---------+---------------------+
10 rows in set (0.00 sec)

mysql> SELECT account_type, COUNT(*) AS total_accounts
    -> FROM accounts
    -> GROUP BY account_type;
+--------------+----------------+
| account_type | total_accounts |
+--------------+----------------+
| SAVINGS      |              6 |
| CURRENT      |              4 |
+--------------+----------------+
2 rows in set (0.01 sec)

mysql> SELECT transaction_type, COUNT(*) AS total_txn
    -> FROM transactions
    -> GROUP BY transaction_type
    -> HAVING COUNT(*) >= 3;
+------------------+-----------+
| transaction_type | total_txn |
+------------------+-----------+
| DEPOSIT          |         3 |
| WITHDRAWAL       |         3 |
| TRANSFER         |         4 |
+------------------+-----------+
3 rows in set (0.01 sec)
```

```
mysql> SELECT u.name, a.account_number, a.balance
    -> FROM users u
    -> INNER JOIN accounts a
    -> ON u.user_id = a.user_id;
+--------------+----------------+----------+
| name         | account_number | balance  |
+--------------+----------------+----------+
| Aarav Kumar  | ACC10001       | 45000.00 |
| Meera Sharma | ACC10002       | 12000.00 |
| Rohan Verma  | ACC10003       | 80000.00 |
| Ananya Iyer  | ACC10004       | 25000.00 |
| Vikram Singh | ACC10005       |  9500.00 |
| Priya Nair   | ACC10006       | 33000.00 |
| Kiran Patel  | ACC10007       | 67000.00 |
| Neha Reddy   | ACC10008       | 15000.00 |
| Sanjay Rao   | ACC10009       |  5000.00 |
| Admin User   | ACC10010       | 99999.00 |
+--------------+----------------+----------+
10 rows in set (0.00 sec)

mysql> SELECT u.user_id, u.name, u.email, a.account_number, a.balance
    -> FROM users u
    -> INNER JOIN accounts a
    -> ON u.user_id = a.user_id;
+---------+--------------+------------------+----------------+----------+
| user_id | name         | email            | account_number | balance  |
+---------+--------------+------------------+----------------+----------+
|       1 | Aarav Kumar  | aarav@gmail.com  | ACC10001       | 45000.00 |
|       2 | Meera Sharma | meera@gmail.com  | ACC10002       | 12000.00 |
|       3 | Rohan Verma  | rohan@gmail.com  | ACC10003       | 80000.00 |
|       4 | Ananya Iyer  | ananya@gmail.com | ACC10004       | 25000.00 |
|       5 | Vikram Singh | vikram@gmail.com | ACC10005       |  9500.00 |
|       6 | Priya Nair   | priya@gmail.com  | ACC10006       | 33000.00 |
|       7 | Kiran Patel  | kiran@gmail.com  | ACC10007       | 67000.00 |
|       8 | Neha Reddy   | neha@gmail.com   | ACC10008       | 15000.00 |
|       9 | Sanjay Rao   | sanjay@gmail.com | ACC10009       |  5000.00 |
|      10 | Admin User   | admin@bank.com   | ACC10010       | 99999.00 |
+---------+--------------+------------------+----------------+----------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT u.user_id, u.name, a.account_number, a.balance
    -> FROM users u
    -> LEFT JOIN accounts a
    -> ON u.user_id = a.user_id;
+---------+--------------+----------------+----------+
| user_id | name         | account_number | balance  |
+---------+--------------+----------------+----------+
|       1 | Aarav Kumar  | ACC10001       | 45000.00 |
|       2 | Meera Sharma | ACC10002       | 12000.00 |
|       3 | Rohan Verma  | ACC10003       | 80000.00 |
|       4 | Ananya Iyer  | ACC10004       | 25000.00 |
|       5 | Vikram Singh | ACC10005       |  9500.00 |
|       6 | Priya Nair   | ACC10006       | 33000.00 |
|       7 | Kiran Patel  | ACC10007       | 67000.00 |
|       8 | Neha Reddy   | ACC10008       | 15000.00 |
|       9 | Sanjay Rao   | ACC10009       |  5000.00 |
|      10 | Admin User   | ACC10010       | 99999.00 |
+---------+--------------+----------------+----------+
10 rows in set (0.00 sec)

mysql> SELECT u.user_id, u.name, a.account_number
    -> FROM users u
    -> RIGHT JOIN accounts a
    -> ON u.user_id = a.user_id;
+---------+--------------+----------------+
| user_id | name         | account_number |
+---------+--------------+----------------+
|       1 | Aarav Kumar  | ACC10001       |
|       2 | Meera Sharma | ACC10002       |
|       3 | Rohan Verma  | ACC10003       |
|       4 | Ananya Iyer  | ACC10004       |
|       5 | Vikram Singh | ACC10005       |
|       6 | Priya Nair   | ACC10006       |
|       7 | Kiran Patel  | ACC10007       |
|       8 | Neha Reddy   | ACC10008       |
|       9 | Sanjay Rao   | ACC10009       |
|      10 | Admin User   | ACC10010       |
+---------+--------------+----------------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     u.name,
    ->     COUNT(t.transaction_id) AS total_transactions
    -> FROM users u
    -> JOIN accounts a ON u.user_id = a.user_id
    -> JOIN transactions t ON a.account_id = t.from_account
    -> GROUP BY u.name;
+--------------+--------------------+
| name         | total_transactions |
+--------------+--------------------+
| Aarav Kumar  |                  1 |
| Meera Sharma |                  1 |
| Rohan Verma  |                  1 |
| Ananya Iyer  |                  1 |
| Priya Nair   |                  1 |
| Kiran Patel  |                  1 |
| Admin User   |                  1 |
+--------------+--------------------+
7 rows in set (0.01 sec)

mysql> SELECT *
    -> FROM users
    -> NATURAL JOIN accounts;
Empty set (0.00 sec)

mysql> SELECT 'users' AS table_name, COUNT(*) AS total FROM users
    -> UNION
    -> SELECT 'accounts', COUNT(*) FROM accounts
    -> UNION
    -> SELECT 'transactions', COUNT(*) FROM transactions
    -> UNION
    -> SELECT 'audit_log', COUNT(*) FROM audit_log;
+--------------+-------+
| table_name   | total |
+--------------+-------+
| users        |    10 |
| accounts     |    10 |
| transactions |    10 |
| audit_log    |    10 |
+--------------+-------+
4 rows in set (0.00 sec)
```